Name: **ANSWER KEY** _____ Date: _____

Inequalities: Color By Number

Color Code:

10 Black $x \le -2$ or $x \ge 2$	**2** Purple $x \ge 3$	**7** Pink $-3 < x \le 3$	**6** Orange $x \le -2$	**9** Red $x < -3$ or $x > 3$
1 Green $x \le -3$	**3** Yellow $x < -3$	**4** Brown $x \ge 2$	**5** Blue $x > 2$	**8** Light Blue $-2 \le x < 3$

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Inequalities: Color By Number

Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.
Solve: $2(x + 3) - 5 \geq 3x + 4$

$2x + 1 \geq 3x + 4$
$-x \geq 3$
$x \leq -3$

1. Color: **Green**

2.
Solve: $\frac{x + 5}{2} \geq 4$

$x + 5 \geq 8$
$x \geq 3$

2. Color: **Purple**

3.
Solve: $-3(x - 1) > 12$

$x - 1 < -4$
$x < -3$

3. Color: **Yellow**

4.
Solve: $5 - 2x \leq 3x - 5$

$10 \leq 5x$
$x \geq 2$

4. Color: **Brown**

5.
Gym A charges a $20 sign-up fee plus $15 per month. Gym B has no sign-up fee and charges $25 per month. Write and solve an inequality to find the number of months, x, for which Gym A costs **less** in total.

$20 + 15x < 25x$
$20 < 10x$
$x > 2$

5. Color: **Blue**

6.
Solve: $6x - 4(x - 1) \leq 0$

$6x - 4x + 4 \leq 0$
$2x \leq -4$
$x \leq -2$

6. Color: **Orange**

7.
Solve: $-5 < 2x + 1 \leq 7$

$-6 < 2x \leq 6$
$-3 < x \leq 3$

7. Color: **Pink**

8.
Solve: $-7 \leq 3x - 1 < 8$

$-6 \leq 3x < 9$
$-2 \leq x < 3$

8. Color: **Light Blue**

9.
Solve: $2x + 1 < -5$ **or** $x - 2 > 1$

$2x < -6$ or $x > 3$
$x < -3$ or $x > 3$

9. Color: **Red**

10.
Solve: $-x + 1 \leq -1$ **or** $3x \leq -6$

$-x \leq -2$ or $x \leq -2$
$x \leq -2$ or $x \geq 2$

10. Color: **Black**